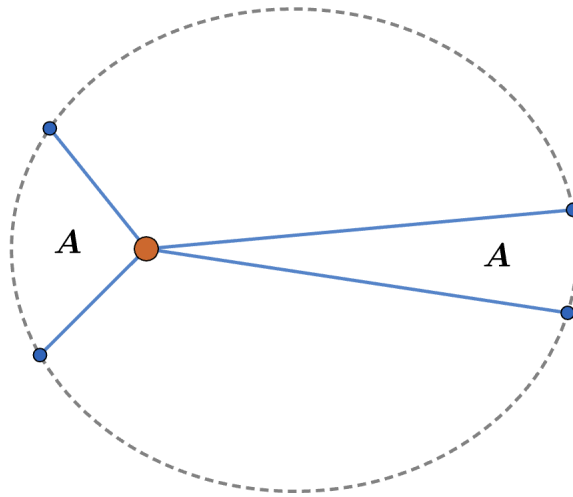


2016 F=ma Exam: Problem 8

Kevin S. Huang



Kepler's laws:

- I) Derived from assuming force $F \propto 1/r^2$, potential energy $V \propto 1/r$
- II) Equivalent to conservation of angular momentum which holds for any central force such as $F \propto 1/r^3$
- III) Derived from assuming force $F \propto 1/r^2$, potential energy $V \propto 1/r$

Thus, the answer is B.